

LESSON 9-1 • HOMEWORK PACKET • LEVEL 1 • SHEET 1 OF 2**Independent and Dependent Variables**

Maryland Standard 6.EE.C.9 • Learning target – I can name the independent and dependent variables in a relationship.

REMEMBER

- The **independent variable** is the input you choose or control.
- The **dependent variable** is the output. It **depends** on the input.
- On a graph: independent on the **x-axis**, dependent on the **y-axis**.

WORKED EXAMPLE

The more cars Andre washes, the more money he earns. Name the variables.

Step 1 Andre controls how many cars he washes.

Step 2 Test: "Money earned depends on cars washed." It makes sense.

Answer Independent: cars washed • Dependent: money earned

WORD BANK

independent variable – the input; what you choose or control

dependent variable – the output; it changes because of the input

depends on – changes because of

x-axis – the line across the bottom of a graph

PRACTICE*show your work • label units*

- 1.** For each pair of quantities, write **I** under the independent variable and **D** under the dependent variable.

Step 1 Read each pair of quantities.

Step 2 Ask: which one changes because of the other? Write D under it.

Step 3 Write I under the other one.

	Quantity	I or D	Quantity	I or D
a.	minutes biking		miles traveled	
b.	total cost		tickets bought	
c.	guests at a party		cups of juice needed	

- 2. Bus Ride.** The longer Imani rides the bus, the farther she gets from home. Write a sentence about these two quantities that uses "depends on."

Frame ____ depends on ____.

Sentence _____

- 3. Dog Walking.** Rosa earns **\$6** for each dog she walks. Name both variables.

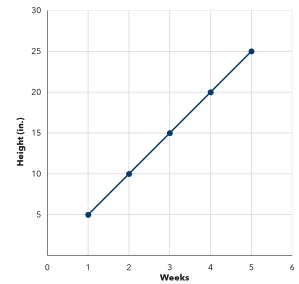
Step 1 Ask: what does Rosa choose?

Step 2 Ask: what changes because of that choice?

Independent / Dependent _____

4. Sunflower. Kenji graphs the height of his sunflower each week.

- a. Name the independent variable. _____
- b. Name the dependent variable. _____
- c. What does the point (4, 20) mean? _____



5. Find the Error. The number of people at a picnic changes how many sandwiches are needed. Leah says the sandwiches are independent because that is the bigger number. Explain her mistake.

Frame Leah is wrong because _____. The independent variable is _____.

CHECK YOURSELF

- ☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____

AT HOME Name two things at home that change together. Which one depends on the other?

LESSON 9-1 • HOMEWORK PACKET • LEVEL 1 • SHEET 2 OF 2**Tables of Values**

Maryland Standard 6.EE.C.9 • Learning target – I can build a table of values for a relationship between two variables.

REMEMBER

- A **table of values** pairs each input (x) with its output (y).
- Use the **rule** on every input. With a fee, **multiply first**, then add.

WORKED EXAMPLE

A copier prints 20 pages each minute. Make a table for 1, 2, and 4 minutes.

Step 1 Minutes is x . Pages is y . The rule is $y = 20x$.**Step 2** $20 \times 1 = 20$, $20 \times 2 = 40$, $20 \times 4 = 80$ **Answer 20, 40, and 80 pages****WORD BANK****table of values** – a list of inputs and their outputs**rule** – what you do to each input**fee** – a cost you pay one time**input** – the number you start with (x)**PRACTICE***show your work • label units*

1. Use the rule
- $y = x + 8$
- to complete the table.

Step 1 Put each x into the rule.**Step 2** Add 8 to get y .

x	0	2	4	7
y				

- 2.
- Skate Rental.**
- A rink charges \$5 to get in plus \$3 for each hour of skating. Complete the table.

Step 1 Multiply the hours by \$3.**Step 2** Then add the \$5 to get in.

Hours (x)	1	2	3	6
Cost in \$ (y)				

- 3.
- Marker Boxes.**
- The table shows markers in boxes. Describe the rule. Then fill in the missing values.

Boxes (x)	1	2	3	4	8
Markers (y)	8	16	24		

Rule _____

- 4.
- Find the Error.**
- Marcus made a table for
- $y = 2x + 1$
- . He wrote
- x
- : 1, 2, 3, 4 and
- y
- : 3, 5, 7, 8. Which
- y
- value is wrong? What should it be?

Frame The wrong value is ____ because $2 \times 4 + 1 =$ ____.**Answer** _____

- 5. Stretch.** A taxi charges \$4 plus \$2 for each mile. Complete the table. How many miles cost \$20?

Step 1 Multiply the miles by \$2, then add \$4.

Step 2 For \$20: take away the \$4, then divide by 2.

Miles (x)	1	3	5	
Cost in \$ (y)				20

TEST PRACTICE

choose the best answer

- 6.** A rule is “multiply by 4, then add 5.” What is y when $x = 6$?

A 24

B 44

C 29

D 15

- 7.** A table has x -values 1, 2, and 3. Which y -values match the rule $y = x + 6$?

A 6, 12, 18

B 7, 8, 9

C 7, 14, 21

D 6, 7, 8

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #____